

Development Summary

Frontend + shared backend — work completed this session

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This document summarizes all the fixes and features implemented for the Fit Force mobile app (Expo / React Native) and its shared backend server during this development session. Items are grouped by area; each entry notes what changed and why.

1. Infrastructure & Cross-Device Data Sync

- **Shared backend server (server/api-server.js, server/db.js)**

The app originally stored everything in per-device AsyncStorage, so a soldier registering on a phone was invisible to an Adjutant approving accounts on a laptop, and messages never reached anyone on a different device. Replaced with a small Node server all devices talk to over the same Wi-Fi as the Expo dev server, backed by a real SQLite database (server/fitforce.db) with proper tables per entity — not just one big JSON blob.

- **Dynamic API base URL (lib/api.ts)**

The client resolves the backend address automatically from the page URL (web) or the Expo dev server host (native), with a configurable fallback for standalone builds — no hardcoded IP, survives Wi-Fi network changes.

- **File upload/download API**

The server accepts base64-encoded file uploads (JSON, no multipart parser needed) and serves them back with correct Content-Type/Content-Disposition, used by Reports (PDF/DOC attachments) and PT Plan PDF export.

- **Server-side PDF generation (server/pdf.js, pdfkit)**

PT plans are rendered into real, text-based PDFs on the server. This replaced expo-print, whose web implementation only opens the browser print dialog (no true "download a file" capability) — the app now does a direct one-tap download on both web and native, no dialog.

- **Dependency/build fixes**

Resolved a recurring "babel-preset-expo not found" failure caused by pnpm's isolated install layout (fixed via --shamefully-hoist), and a React/react-native-renderer version mismatch that crashed the app on native devices immediately after load.

2. Authentication & Registration

- **Medical category simplified**

Replaced the old "SHAPE-1/2/3 / Temporarily Unfit" options with Category A / B / C across registration and profile.

- **Date of Birth replaces manual Age entry**

Registration now collects date of birth (validated) and the app computes age automatically against the current date, instead of trusting a typed-in number.

- **Address field added**

Required on registration, editable on profile, shown on the soldier detail view.

- **Previous IPFT result capture**

New soldiers can record a prior unit's IPFT result (Pass/Fail) and which biannual cycle it was from. If provided, it's saved as a real, clearly-labeled entry ("Self-reported at registration") in the soldier's IPFT history — informational only, does not affect their pending status at the new unit.

- **Critical bug: registration form losing keyboard focus every keystroke**

Input/dropdown components were defined inside the screen's render function, so every keystroke recreated them as new component types and React remounted the text field, dropping focus after each character. Fixed by hoisting them outside the component so their identity stays stable across renders.

- **Pending-approval workflow**

Self-registered soldiers stay inactive until an Adjutant approves them; Adjutant can also delete a pending registration outright.

3. Profile Management

- **Edit Profile added for every role**

Previously view-only; now editable in place with change tracking.

- **Field set reworked**

Removed Blood Group and Medical Category from self-edit (clinician-verified fields, not self-reported); added Email (with uniqueness check), Phone, Company, Height, Weight, Age, Address.

- **Soldier edits notify the Clerk**

When a soldier updates their profile, the Clerk receives a notification naming exactly which fields changed (e.g. "age, weight, address").

4. IPFT Testing & Results

- **Critical bug: run times always scored FAIL**

The result form tried to parse free-typed run times as an exact "MM:SS" string and silently marked the item FAIL whenever the parse didn't match — which was most of the time. Removed the parsing entirely.

- **Pass/Fail is now a direct tap per item**

Every IPFT item (including the two run distances) is graded with explicit PASS/FAIL buttons instead of inferring it from typed text, with an optional free-text note field for the record. This matches how these items are actually graded on the ground.

- **Run distances corrected**

Replaced the single 2.4km run with the two required distances: 1.6km (pass under 9 min) and 3.2km (pass under 24 min).

- **Obstacle-course items added**

Swimming, Beam, Knee Bending, Horizontal Rope, Fireman's Lift, and Wall are now part of the recordable IPFT item set alongside Push-ups, Sit-ups and Pull-ups.

- **Result notification names the weak items**

A soldier failing their test now gets a notification listing exactly which items they were weak in, not just an overall Pass/Fail.

- **Fit/Unfit status leads straight to the correct PT plan**

Saving a result immediately flips the soldier's status and their dashboard shows the matching plan (Fit vs. Rehabilitation) with no manual step.

- **Full history preserved**

Every result is appended, never overwritten — a soldier's complete IPFT history stays visible on their profile.

- **Biannual tagging**

Results can be tagged 1st/2nd biannual cycle, shown as a badge in the history list.

- **IPFT date scheduling**

Adjutant proposes a date and CO approves/rejects it using a real tappable calendar (built in-house, no extra native dependency) instead of typed-in date strings.

5. PT Plans

- **Fit and Rehabilitation plan types**

Soldiers are shown the plan matching their current status; 2IC creates/publishes plans for both.

- **True one-tap PDF download**

Fixed to download a real PDF file directly (server-generated, via pdfkit) instead of opening a print dialog on web or requiring extra steps on native.

6. Messaging, Notifications & Reports

- **Messaging separated from official workflows**

Messages is now plain free-text chat between staff roles (everyone except Soldiers) — it no longer carries IPFT approval logic or file attachments.

- **Notification targeting bug (privacy leak)**

Company-wide staff alerts (e.g. "Soldier X is overweight") were visible to every soldier in that company, including peers of the soldier being discussed. Fixed: soldiers only ever see notifications addressed specifically to them, or true all-unit broadcasts with no targeting at all.

- **Reports: real file attachments**

Clerk selects a specific recipient (CO / 2IC / Coy Comd / Adjutant) first, then can attach a real PDF or Word document via a file picker — uploaded to the server and downloadable by the recipient, replacing the earlier text-only "attachment label".

- **Screens refresh on focus**

Messages, Notifications, Reports, Dashboard and Soldiers now refetch every time you switch to that tab,

instead of only on first mount — so updates from other devices actually show up.

7. Dashboard & Soldier Roster

- **IPFT result graphs**

Company pass-rate chart (CO), Fit/Unfit/Overweight chart (Company Commander), and a personal trend chart (Soldier) — built as a lightweight custom chart component, no new dependency.

- **"Total Strength" not updating after approval**

A soldier's record is created the moment they register, so they were already counted in "Total" before approval — meaning approving them caused no visible change. Fixed so Total/Fit/Unfit/Overweight counts only include approved soldiers.

- **Approve/Delete for pending registrations**

Adjutant can approve or permanently delete a pending registration directly from the roster, with instant UI feedback.

- **Delete any soldier**

Adjutant can also delete an already-approved soldier's account from the roster, with confirmation.

- **Clearer status labels**

The "PENDING" filter chip (IPFT test status) was easily confused with registration-pending; relabeled to "AWAITING TEST".

8. Cross-Cutting Fixes

- **Critical bug: confirm dialogs silently did nothing on web**

react-native-web's `Alert.alert()` is a complete no-op in the browser — no dialog appears and no button's `onPress` ever fires. This silently broke every confirm/success/error dialog across the app on web, including Approve, Delete, and navigating back after a successful save. Built a cross-platform `showAlert()` helper (falls back to `window.alert/confirm` on web, keeps the native dialog on phone) and replaced every `Alert.alert` call across all affected screens.

- **Consistent theme across devices**

The app previously followed each device's own OS dark/light setting, so a phone and laptop could show different palettes. Forced a single consistent theme everywhere.

- **Larger, clearer logo and icons**

Enlarged the header logo, registration screen logo, and tab bar icons for legibility.

- **Correct QR code / LAN links**

Generated and kept in sync with the current network address for Expo Go and browser access from other devices.

End of summary.